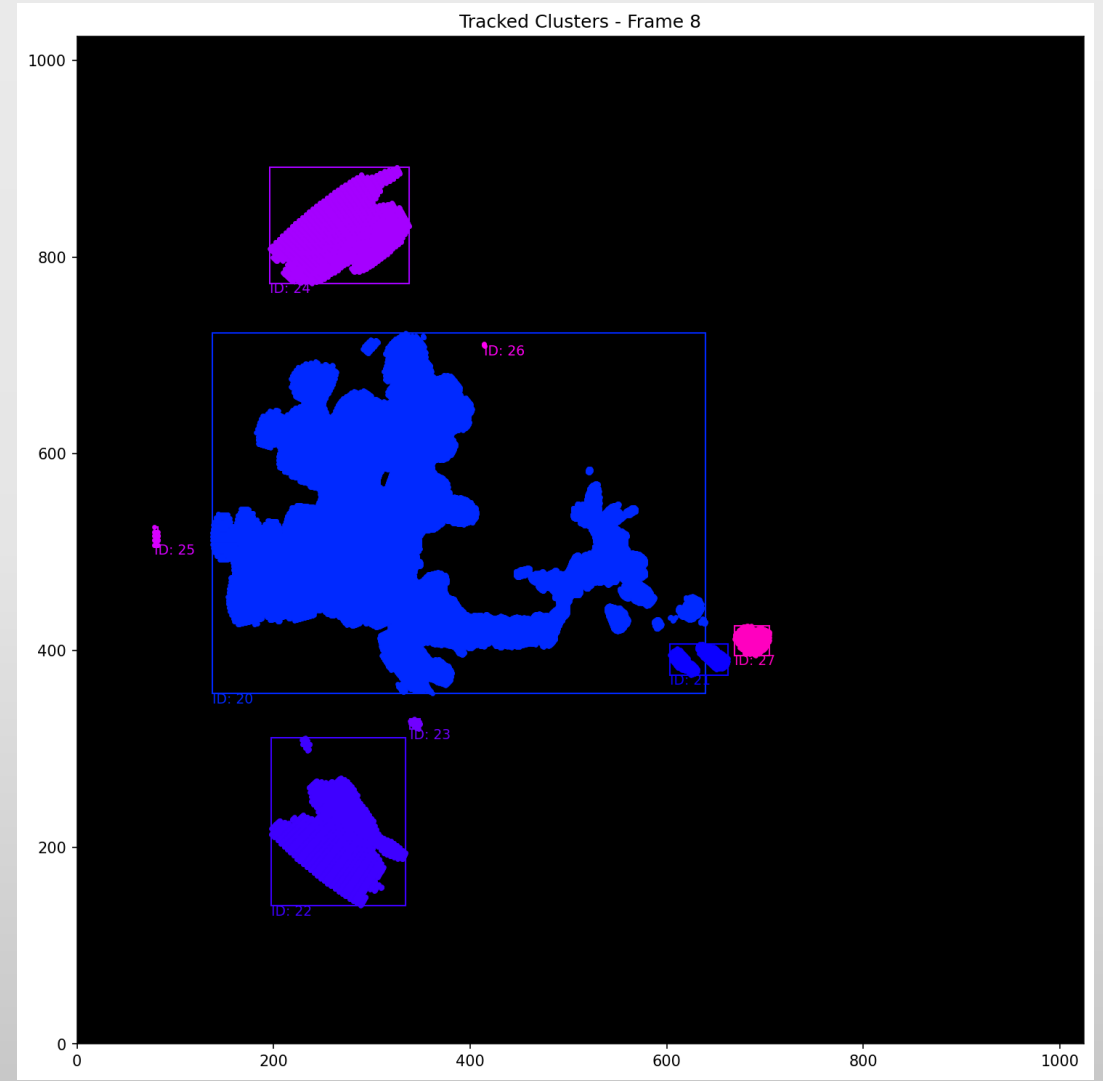


STDBSCAN

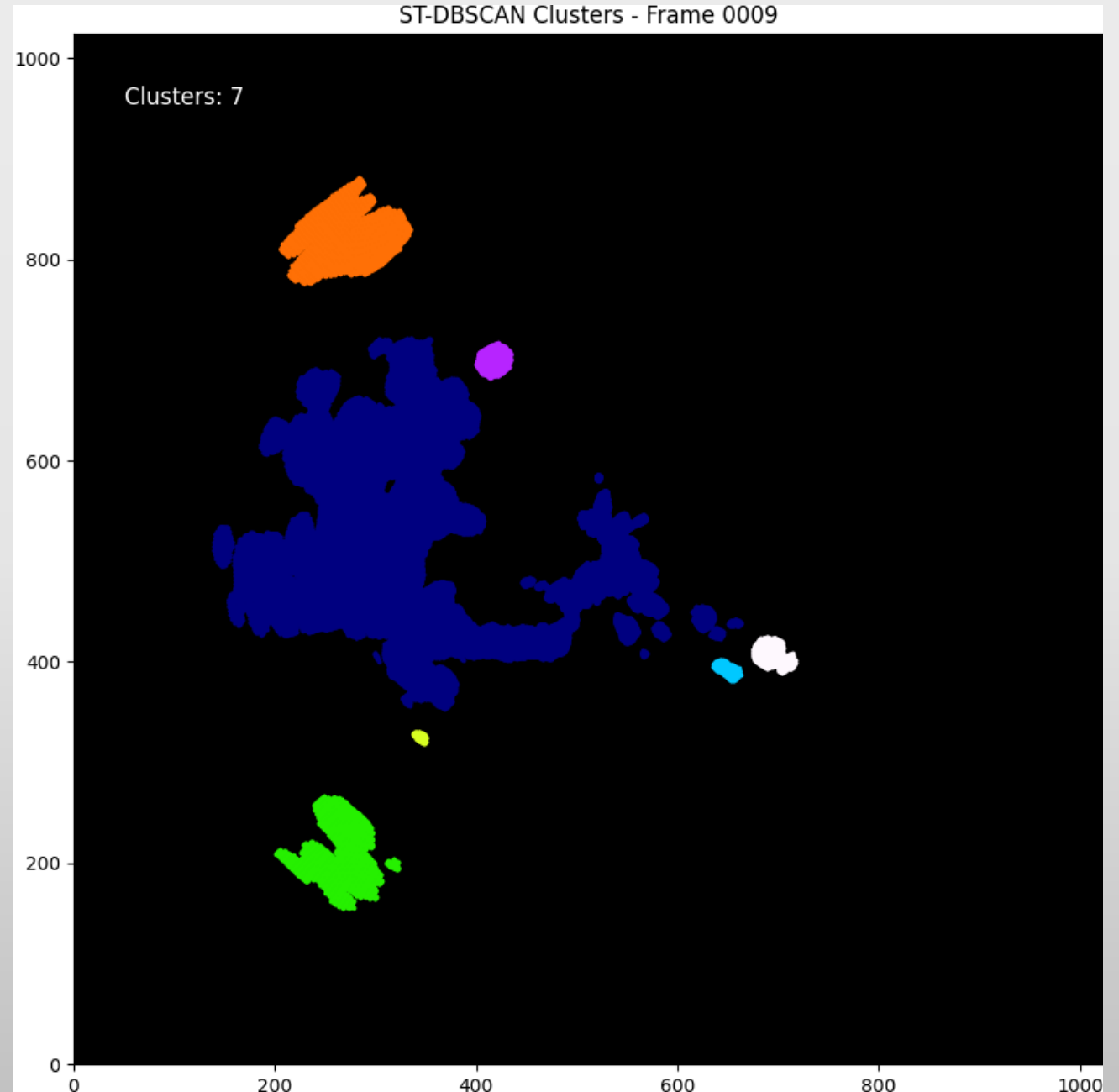
Sam Cancilla 9-30-2025



STDBSCAN

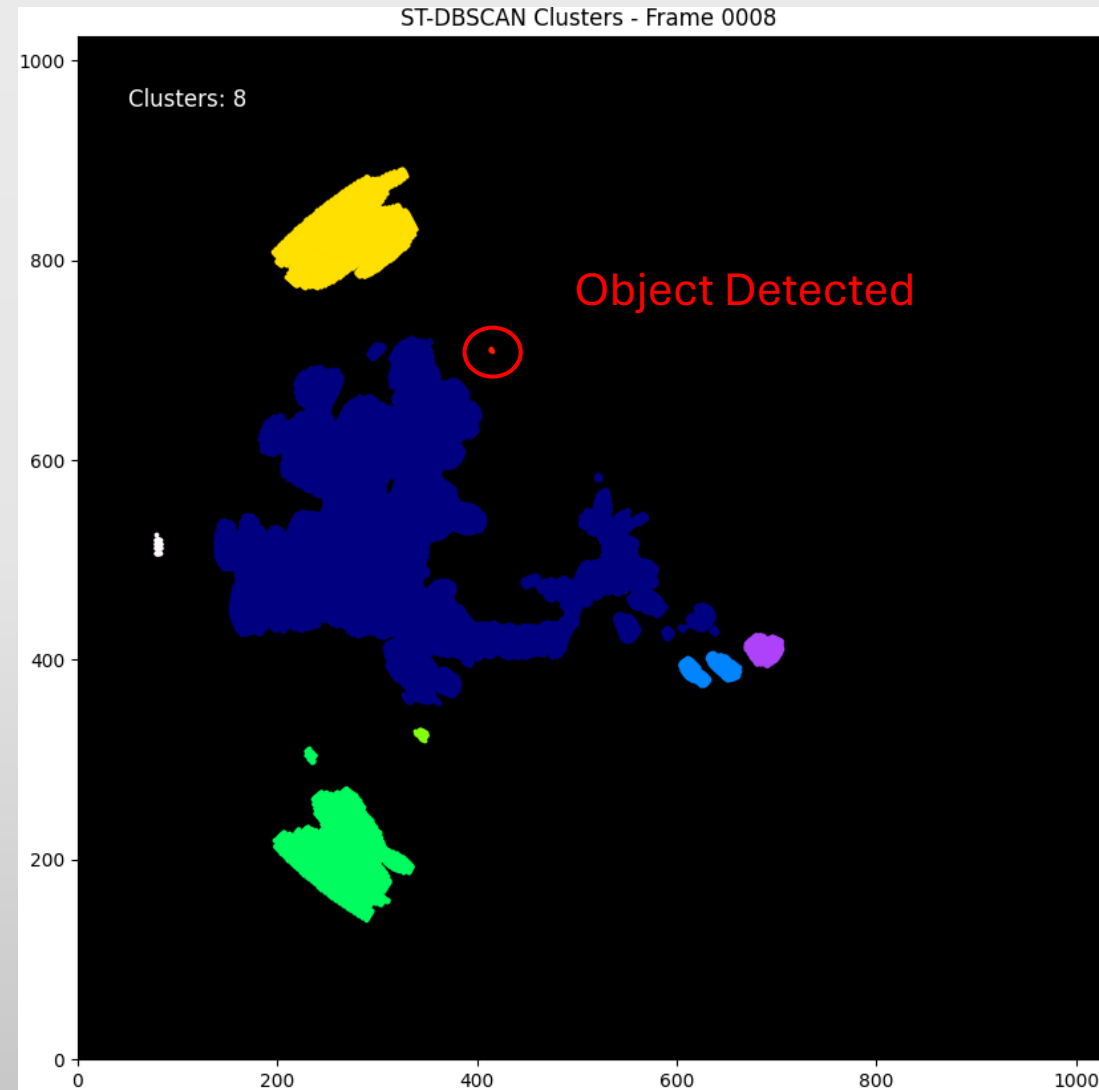
ST-DBSCAN

- Spatio-Temporal
 - (x, y, t)
- Parameters are all the same as DBSCAN
 - Exception: ϵ_1 (spatial), ϵ_2 (temporal)
- Better for evolving datasets (movement)
- Incredibly taxing and time consuming to process data



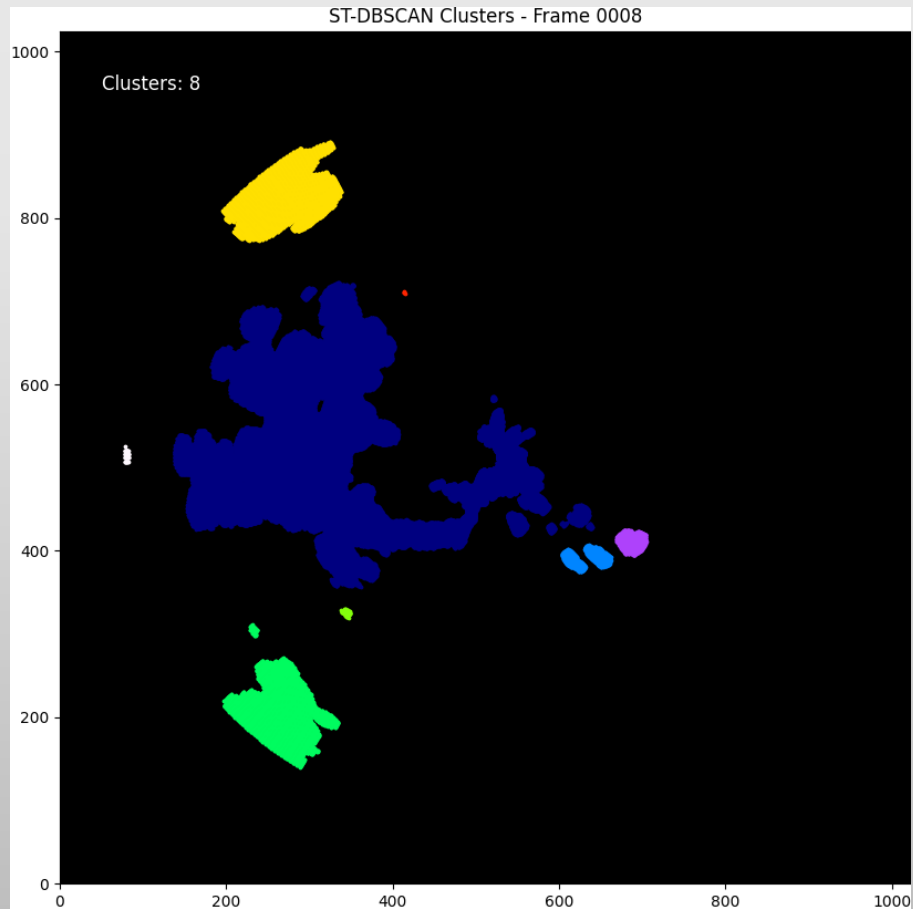
Data with ST-DBSCAN

- There are few images due to processing time
 - A couple hours for 10 images
- It is working when compared to raw images
- Could probably shave down time to process with optimization



Comparison

ST-DBSCAN



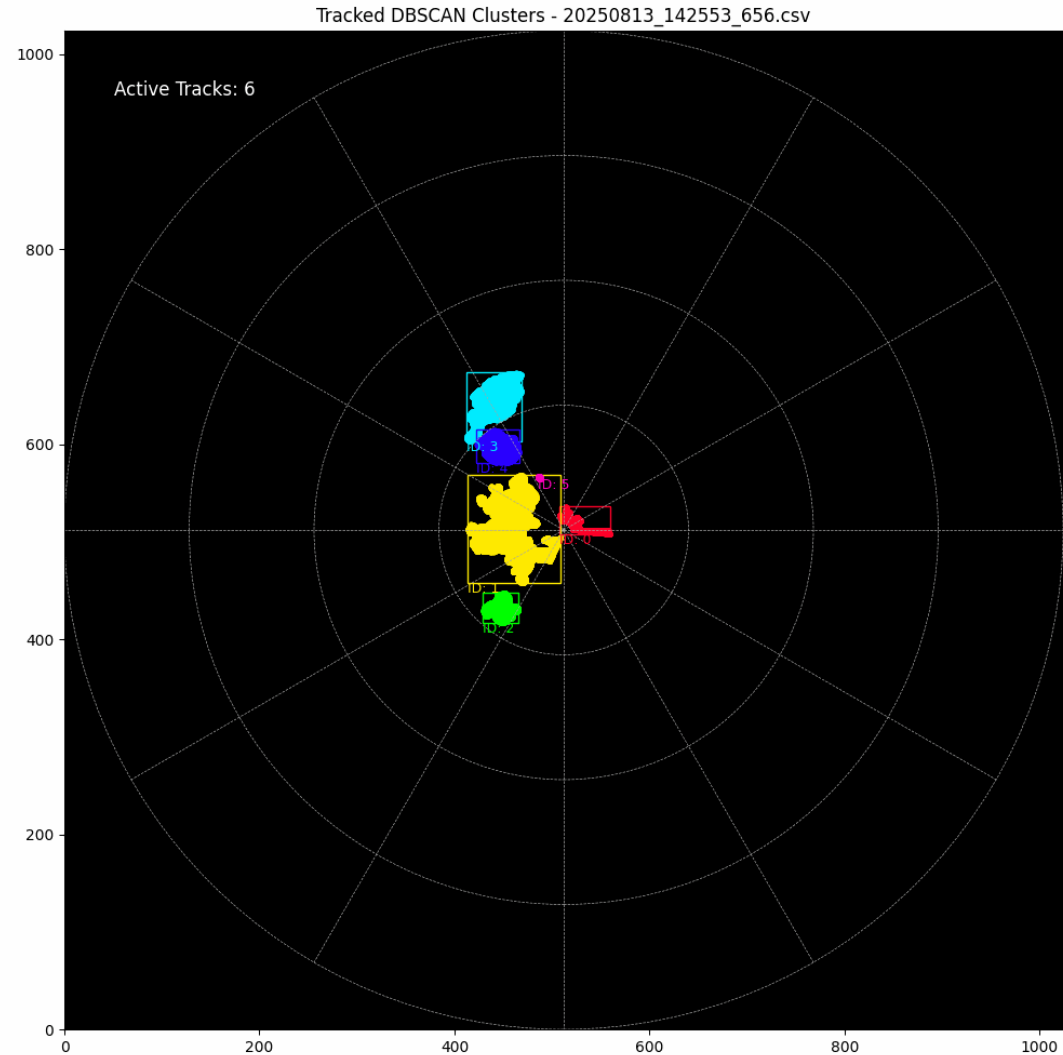
Raw Image



Tracking Implementation

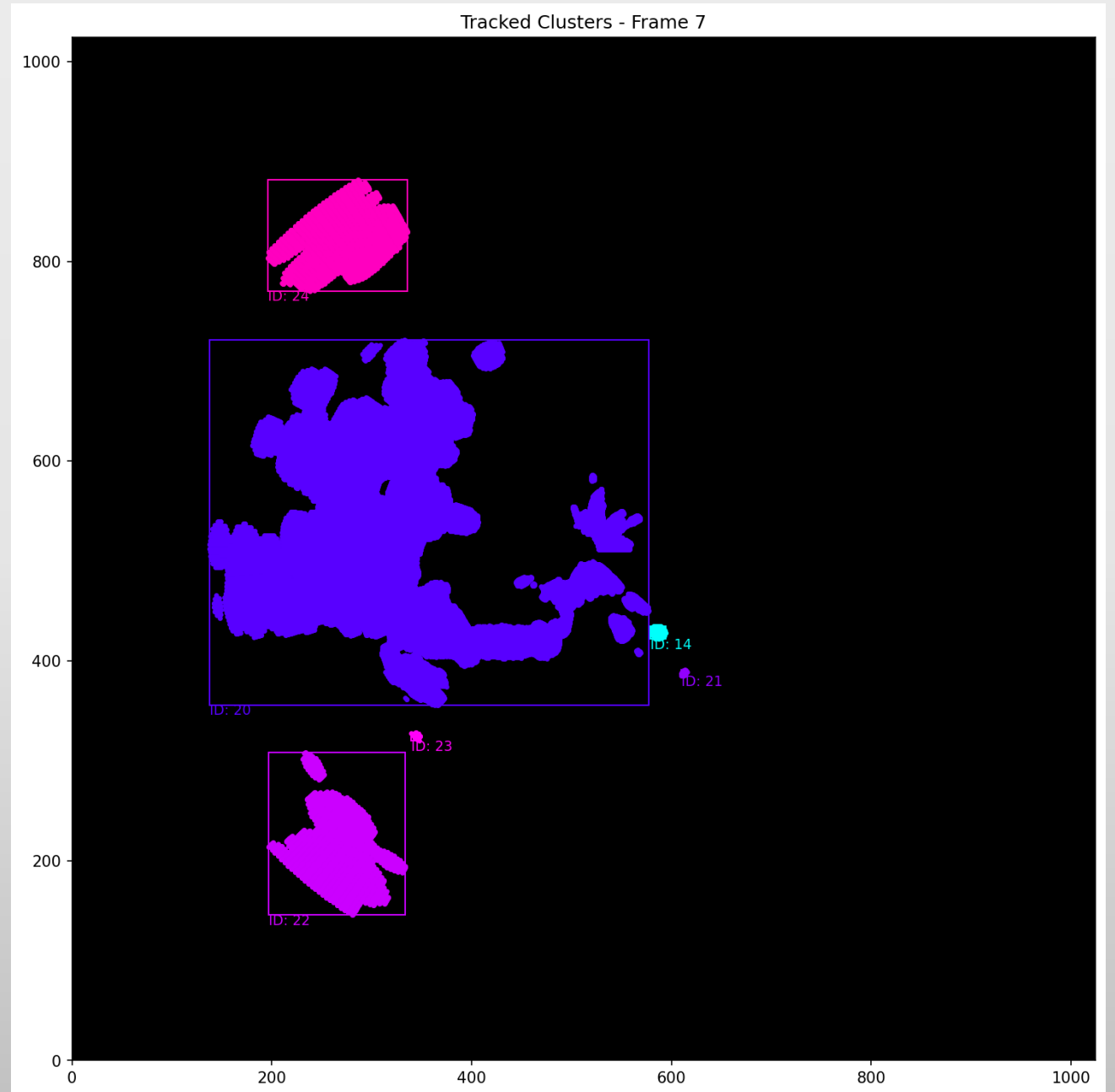
Tracking DBSCAN

- Used strict parameters to ensure that objects on the water did not get mixed with land
- Can clearly see the object is tracked
 - Unfortunately does blip because not ST-DBSCAN
- The tracking and labeling mechanism does work



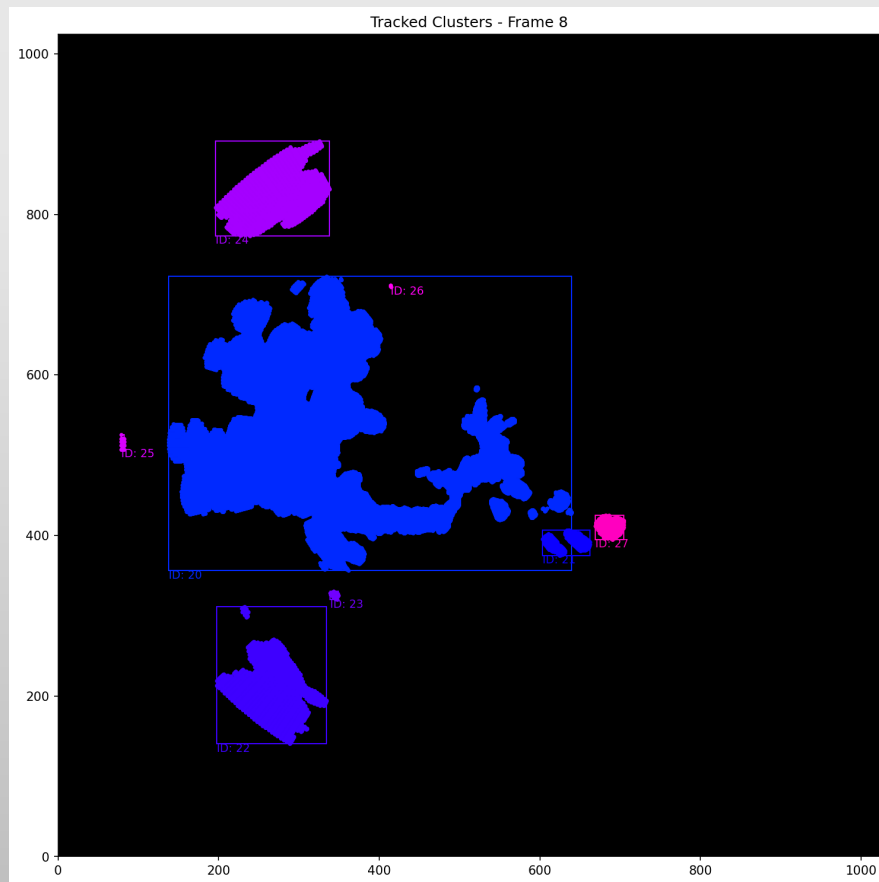
Tracked ST-DBSCAN Frame

- Will need to apply stricter filtering
 - Sea object is apart of land cluster
- Can see that objects are being tracked
- The IDs stay when tracking objects over time

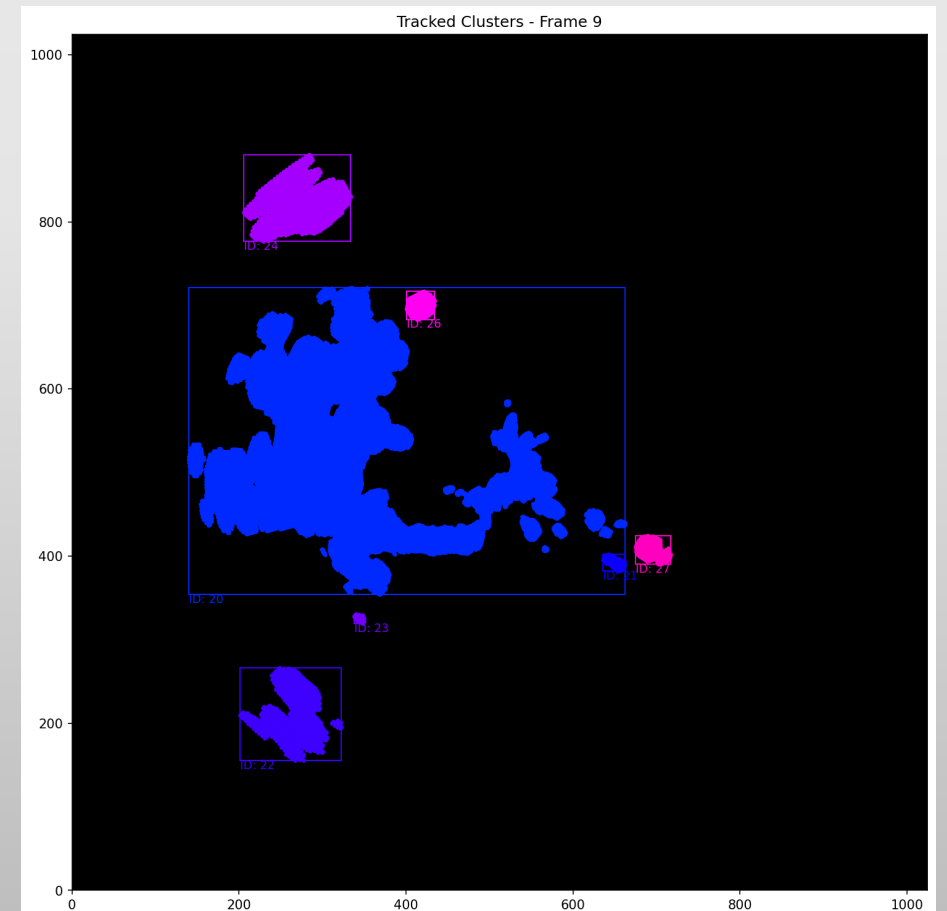


Comparison

ST-DBSCAN Blip fill in



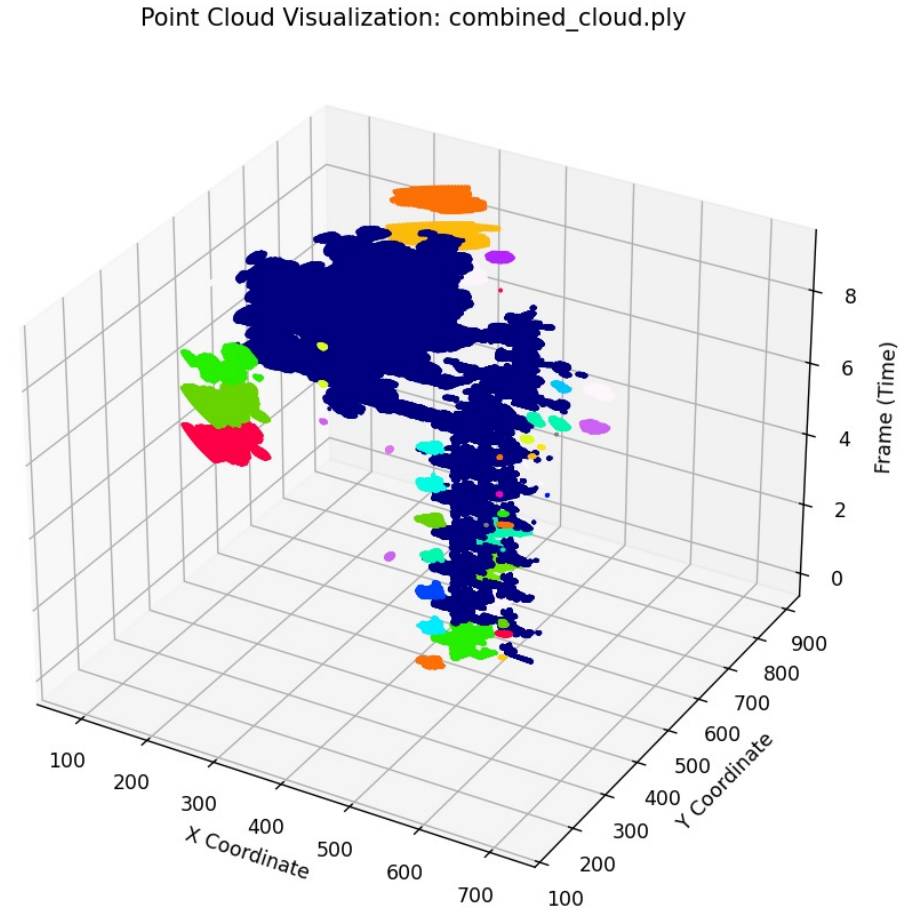
ST-DBSCAN Frame after



Point Cloud

Point Cloud

- Prompted Gemini last night
 - Gemi took several images and layered them on top of one another
- Not really sure what we are looking for
 - Maybe a strong point is seen as 'heightened' while a weak point is 'sunken'



Moving forward

- Process more ST-DBSCAN frames
 - Optimization
- Point Cloud on a single frame



The background is a vibrant blue with a repeating pattern of colorful speech bubbles. Each bubble is a different color (red, yellow, pink, white) and contains a large, bold, dark blue question mark. The bubbles are scattered across the entire frame, creating a sense of many questions.

Thank you.

Any Questions?